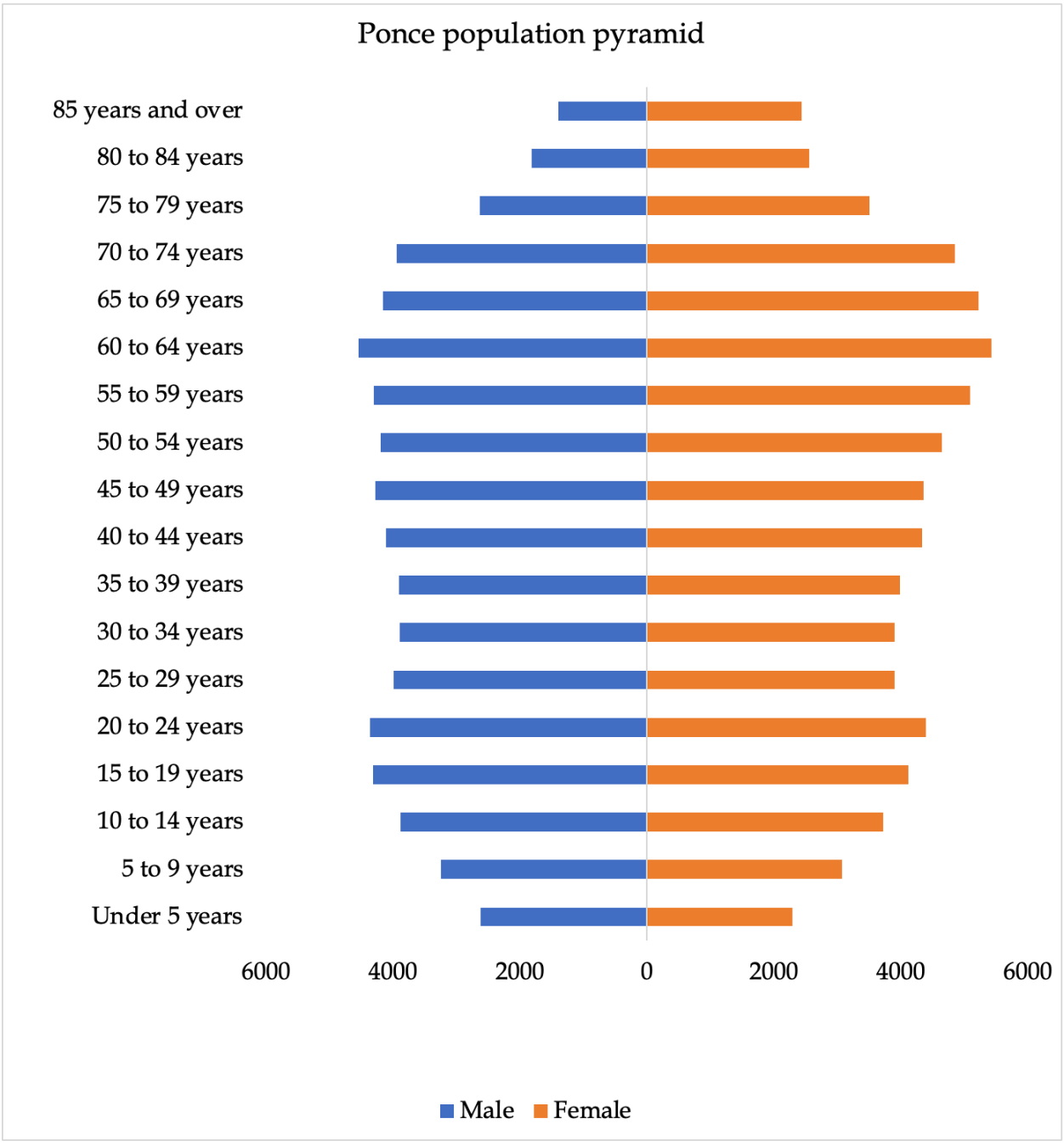
**Julian Salcedo**

**Escuela Graduada de Planificación**

**Análisis Demográfico del Municipio de Ponce**

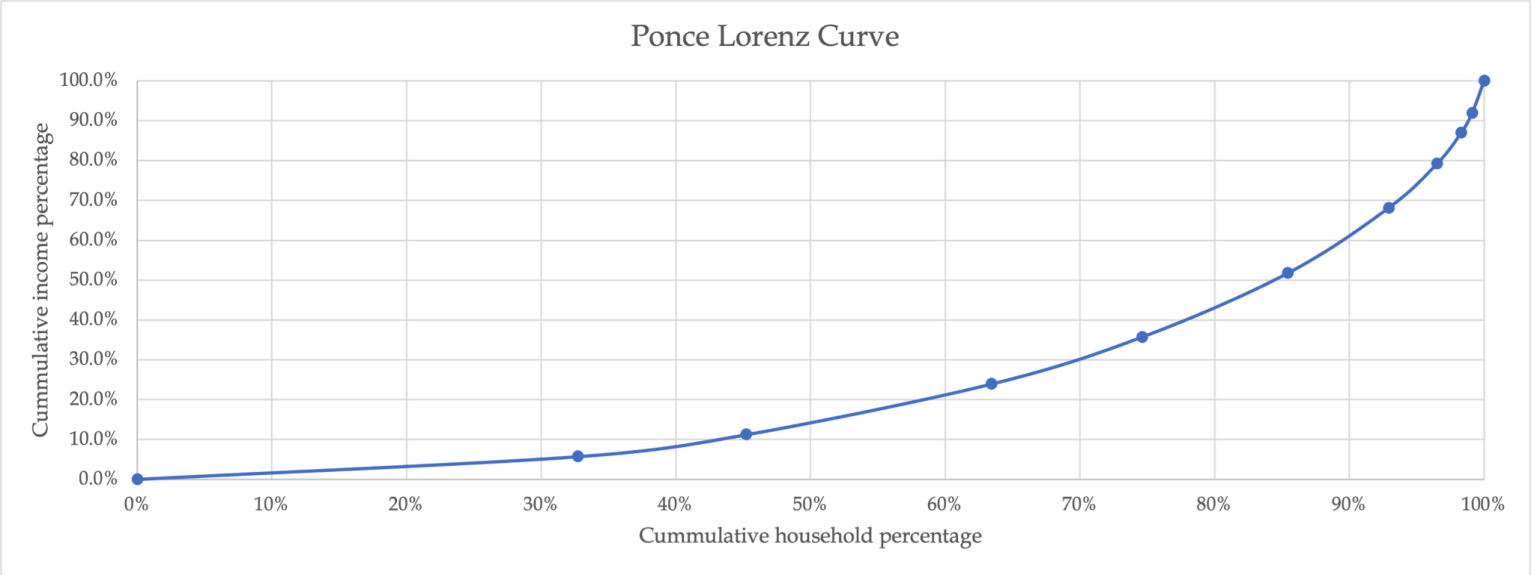
La pirámide poblacional de Ponce presenta baja natalidad (El segmento más pequeño es el de menos de 5 años). Los cohortes más grandes van de los 50-70 años. De los 15-40 años están los cohortes más reducidos después de los niños. También puede inferirse una fuerza laboral reducida. No hay niños naciendo para reemplazar la fuerza laboral. Se puede inferir con esta piramide poblacional que es una población que está envejeciendo.


Ponce population pyramid
85 years and over
80 to 84 years
75 to 79 years
70 to 74 years
65 to 69 years
60 to 64 years
55 to 59 years
50 to 54 years
45 to 49 years
40 to 44 years
35 to 39 years
30 to 34 years
25 to 29 years
20 to 24 years
15 to 19 years
10 to 14 years
5 to 9 years
Under 5 years
6000
4000
2000
0
2000
4000
6000
Male
Female

**Análisis de Ingresos del Municipio de Ponce**

La Curva de Lorenz muestra una alta desigualdad en el municipio de Ponce. Se puede ver como se separa la linea y forma una curvatura que indica la desigualdad. El 85% de los hogares acumula el 51% de los ingresos. El 15% de los hogares acumula el 49% del total de los ingresos. Esto implica una desigualdad muy alta en el Municipio de Ponce. Aproximadamente 7600 hogares -de un total de 52 mil hogares- acumulan 723 millones. El total de ingresos de todas las familias es de 1.4 billones.

| | | |
|---|---|---|
| Total households | | 52439 |
| Mean income | $ | 28,580.00 |
| Total income | $ | 1,498,706,620.00 |



Los cambios a nivel demográfico en los barrios presentan una reducción poblacional sumado al envejecimiento poblacional para la mayoría de los barrios. Un 87% de los barrios tuvo una disminución poblacional.Sólo un 13% aumentó su población. En cuanto a la mediana de edad, 27 de 31 barrios tuvieron un aumento.  A nivel socioeconómico, entre 2019 y 2024 un 29% de los barrios aumentó su cantidad de personas bajo el nivel de pobreza. En cuanto a la fuerza laboral, un 83% de los barrios tuvo una reducción de la cantidad de personas en la fuerza laboral y solo en un 16% aumento la misma. Un 64% de los barrios aumentaron su población en desempleo.

Las implicaciones dado este análisis y el llevado a cabo en la pregunta anterior indican que se debe desarrollar una planificación que contemple estas realidades de reducción poblacional y envejecimiento, asi como un aumento en el desempleo y reducción de la fuerza laboral. Esto implicaría el desarrollo de alternativas a la planificación basada en el crecimiento por una planificación basada en la contracción o decrecimiento que sirva estas poblaciones.

Tabla 1. Bajo Nivel de Pobreza

| Barrio | Under Poverty 2014 | Under Poverty 2019 | Under Poverty 2024 |
|---|---|---|---|
| Canas | 15,264 | 12,455 | 11,773 |
| Canas Urbano | 8,707 | 7,702 | 7,239 |
| Playa | 7,612 | 5,440 | 5,554 |
| Machuelo Abajo | 5,012 | 5,710 | 5,409 |
| Machuelo | 5,481 | 5,278 | 4,682 |

| | | | |
|---|---|---|---|
| Arriba | | | |
| San Antón | 5,889 | 5,128 | 4,012 |
| Segundo | 5,243 | 3,210 | 3,441 |
| Coto Laurel | 2,541 | 2,457 | 3,025 |
| Sabanetas | 2,293 | 2,539 | 2,581 |
| Portugués | 3,236 | 3,032 | 2,340 |
| Primero | 2,185 | 2,048 | 1,873 |
| Magueyes | 1,750 | 1,628 | 1,640 |
| Sexto | 2,337 | 2,109 | 1,633 |
| Portugués Urbano | 2,456 | 1,911 | 1,466 |
| Real | 1,463 | 1,454 | 962 |
| Bucana | 779 | 562 | 838 |
| Cerrillos | 1,116 | 1,272 | 818 |
| Marueño | 730 | 974 | 687 |
| Anón | 1,192 | 742 | 558 |
| Cuarto | 988 | 839 | 552 |
| Capitanejo | 570 | 792 | 527 |
| Quinto | 199 | 374 | 483 |
| Quebrada Limón | 461 | 165 | 466 |
| Guaraguao | 405 | 267 | 459 |
| Magueyes Urbano | 324 | 393 | 322 |
| Tercero | 685 | 440 | 316 |
| Tibes | 595 | 366 | 234 |
| Vayas | 250 | 256 | 176 |
| Maragüez | 223 | 299 | 128 |
| San Patricio | 455 | 277 | 116 |
| Monte Llano | 120 | 144 | 65 |

Fuente: US Census ACS

Tabla 2. Fuerza Laboral

| Barrio | Labor Force 2014 | Labor Force 2019 | Labor Force 2024 |
|---|---|---|---|
| Canas | 24,348 | 22,604 | 21,825 |

| | | | |
|---|---|---|---|
| Canas Urbano | 14,354 | 13,044 | 13,391 |
| Playa | 10,835 | 9,559 | 9,914 |
| Machuelo Abajo | 8,279 | 8,552 | 8,577 |
| Machuelo Arriba | 9,553 | 9,274 | 8,540 |
| San Antón | 8,068 | 7,855 | 7,227 |
| Coto Laurel | 4,966 | 4,137 | 6,767 |
| Sabanetas | 3,895 | 4,345 | 4,507 |
| Magueyes | 5,185 | 5,474 | 4,405 |
| Segundo | 5,940 | 3,907 | 4,276 |
| Cerrillos | 4,142 | 3,316 | 3,070 |
| Portugués | 3,027 | 2,743 | 2,666 |
| Bucana | 2,997 | 2,365 | 2,336 |
| Portugués Urbano | 3,105 | 2,573 | 2,163 |
| Primero | 2,402 | 2,029 | 2,086 |
| Sexto | 2,930 | 2,203 | 1,906 |
| Real | 2,854 | 2,140 | 1,725 |
| Cuarto | 1,217 | 1,530 | 1,074 |
| Marueño | 1,116 | 1,418 | 1,035 |
| Magueyes Urbano | 840 | 885 | 895 |
| Anón | 1,378 | 944 | 690 |
| Quinto | 443 | 487 | 607 |
| Vayas | 613 | 529 | 591 |
| Quebrada Limón | 598 | 385 | 584 |
| Capitanejo | 656 | 792 | 558 |
| Tercero | 599 | 441 | 503 |
| Guaraguao | 985 | 408 | 474 |
| Tibes | 768 | 367 | 303 |
| Maragüez | 297 | 319 | 190 |
| Monte Llano | 336 | 365 | 177 |
| San Patricio | 419 | 329 | 177 |

Tabla 3. En la Fuerza Laboral

| Barrio | Employed 2014 | Employed 2019 | Employed 2024 |
|---|---|---|---|
| Canas | 8,109 | 7,688 | 7,445 |
| Canas Urbano | 4,705 | 4,665 | 4,720 |
| Playa | 3,896 | 2,802 | 3,673 |
| Coto Laurel | 2,230 | 1,393 | 3,200 |
| Machuelo Arriba | 3,147 | 3,350 | 2,838 |
| Machuelo Abajo | 2,765 | 2,380 | 2,559 |
| San Antón | 2,805 | 2,412 | 2,265 |
| Sabanetas | 1,184 | 1,254 | 1,622 |
| Magueyes | 2,141 | 2,536 | 1,420 |
| Segundo | 1,621 | 1,023 | 1,245 |
| Cerrillos | 1,676 | 1,612 | 1,167 |
| Bucana | 1,373 | 1,127 | 981 |
| Portugués | 770 | 623 | 829 |
| Real | 1,189 | 902 | 768 |
| Primero | 627 | 415 | 727 |
| Portugués Urbano | 1,024 | 638 | 604 |
| Sexto | 853 | 353 | 513 |
| Cuarto | 445 | 439 | 410 |
| Magueyes Urbano | 269 | 245 | 372 |
| Marueño | 412 | 385 | 332 |
| Quebrada Limón | 156 | 139 | 291 |
| Vayas | 165 | 114 | 223 |
| Guaraguao | 316 | 63 | 210 |
| Anón | 425 | 255 | 155 |
| Capitanejo | 125 | 130 | 125 |
| Tibes | 257 | 109 | 111 |
| Monte Llano | 205 | 142 | 91 |
| Tercero | 126 | 159 | 91 |
| San Patricio | 27 | 113 | 58 |
| Quinto | 76 | 204 | 54 |
| Maragüez | 56 | 90 | 33 |

Fuente: US Census ACS

Tabla 4. Personas Desempleadas

| Barrio | Unemployed 2014 | Unemployed 2019 | Unemployed 2024 |
|---|---|---|---|
| Canas Urbano | 1,397 | 732 | 825 |
| Machuelo Abajo | 627 | 322 | 756 |
| Coto Laurel | 592 | 233 | 696 |
| Machuelo Arriba | 906 | 413 | 617 |
| San Antón | 936 | 443 | 521 |
| Portugués | 481 | 152 | 484 |
| Segundo | 622 | 306 | 471 |
| Playa | 1,116 | 379 | 463 |
| Canas | 2,412 | 897 | 435 |
| Sabanetas | 307 | 196 | 255 |
| Sexto | 378 | 173 | 228 |
| Portugués Urbano | 498 | 110 | 208 |
| Quinto | 43 | 21 | 167 |
| Magueyes | 275 | 115 | 159 |
| Tercero | 72 | 39 | 95 |
| Cuarto | 142 | 121 | 91 |
| Primero | 249 | 171 | 91 |
| Anón | 153 | 0 | 78 |
| Real | 347 | 29 | 77 |
| Cerrillos | 183 | 129 | 71 |
| Bucana | 268 | 148 | 29 |
| Maragüez | 31 | 0 | 29 |
| Monte Llano | 11 | 0 | 29 |
| Vayas | 46 | 0 | 26 |
| Capitanejo | 92 | 47 | 22 |
| Magueyes Urbano | 90 | 72 | 18 |
| Tibes | 35 | 0 | 18 |
| Guaraguao | 103 | 0 | 0 |
| Marueño | 70 | 36 | 0 |
| Quebrada Limón | 78 | 14 | 0 |
| San Patricio | 19 | 33 | 0 |

Fuente: US Census ACS

Tabla 5. Mediana de edad

| Barrio | Age 2014 | Age 2019 | Age 2024 |
|---|---|---|---|
| Quinto | 58.7 | 54.5 | 61.3 |
| Vayas | 57.9 | 60.3 | 60.3 |
| Maragüez | 47.8 | 42.7 | 57.7 |
| Cerrillos | 41.6 | 37.2 | 56.2 |
| San Patricio | 46.1 | 36.4 | 56.2 |
| Magueyes | 41.1 | 39.9 | 53.5 |
| Anón | 37.9 | 43.9 | 53.1 |
| Portugués Urbano | 33.1 | 44.2 | 51.9 |
| Machuelo Arriba | 40.1 | 43.3 | 49.9 |
| Machuelo Abajo | 39.4 | 41.9 | 49.4 |
| Cuarto | 40.9 | 47.6 | 49.1 |
| Segundo | 41.9 | 40.5 | 49.0 |
| Capitanejo | 45.1 | 43.1 | 47.9 |
| Monte Llano | 44.4 | 53.2 | 47.2 |
| San Antón | 33.5 | 40.4 | 47.2 |
| Marueño | 37.0 | 43.5 | 47.1 |
| Bucana | 44.3 | 50.0 | 46.2 |
| Sabanetas | 41.9 | 49.3 | 45.2 |
| Sexto | 38.1 | 39.3 | 43.0 |
| Playa | 36.5 | 44.4 | 42.6 |
| Magueyes Urbano | 50.9 | 50.0 | 42.5 |
| Coto Laurel | 39.8 | 44.4 | 41.4 |
| Canas Urbano | 38.4 | 39.7 | 41.2 |
| Portugués | 29.1 | 34.5 | 41.1 |
| Tercero | 33.4 | 26.9 | 41.0 |
| Real | 37.4 | 30.1 | 40.5 |
| Primero | 34.7 | 31.6 | 40.2 |
| Canas | 34.1 | 39.3 | 39.3 |
| Guaraguao | 40.1 | 33.5 | 38.7 |
| Tibes | 39.7 | 28.8 | 38.4 |
| Quebrada Limón | 44.0 | 53.4 | 28.0 |

Fuente: US Census ACS

Tabla 6. Mediana de Ingresos

| Barrio | Income 2014 | Income 2019 | Income 2024 |
|---|---|---|---|
| Cerrillos | $37,574 | $30,933 | $37,370 |
| Magueyes Urbano | $21,283 | $33,241 | $33,380 |
| Magueyes | $30,293 | $35,000 | $31,844 |
| Bucana | $31,693 | $27,409 | $30,968 |
| Coto Laurel | $22,133 | $17,664 | $26,809 |
| Canas | $17,454 | $19,274 | $23,782 |
| Real | $20,514 | $20,754 | $22,083 |
| Guaraguao | $24,306 | $10,878 | $22,000 |
| Sabanetas | $18,039 | $17,882 | $21,919 |
| Machuelo Arriba | $19,850 | $20,995 | $21,837 |
| Playa | $16,332 | $17,266 | $19,740 |
| Canas Urbano | $15,587 | $17,646 | $19,294 |
| Portugués Urbano | $15,875 | $14,028 | $17,689 |
| Sexto | $10,798 | $9,626 | $17,025 |
| Marueño | $14,505 | $16,953 | $17,000 |
| San Antón | $17,561 | $13,507 | $16,720 |
| Cuarto | $11,147 | $12,127 | $16,058 |
| Machuelo Abajo | $16,914 | $15,575 | $15,858 |
| Tibes | $17,628 | $11,250 | $15,673 |
| Capitanejo | $10,298 | $13,320 | $14,821 |
| Maragüez | $13,629 | $9,800 | $14,620 |
| Anón | $10,739 | $11,647 | $13,808 |
| Segundo | $9,929 | $10,852 | $12,059 |
| Portugués | $5,871 | $7,120 | $11,151 |
| Quinto | $14,257 | $12,429 | $10,080 |
| Primero | $8,228 | $7,776 | $8,817 |
| Monte Llano | $32,188 | $0 | $0 |
| Quebrada Limón | $14,012 | $0 | $0 |
| San Patricio | $7,538 | $0 | $0 |
| Tercero | $9,833 | $11,594 | $0 |

| | | | |
|---|---|---|---|
| Vayas | $16,442 | $14,076 | $0 |

Fuente: US Census ACS

Tabla 7. Población total

| Barrio | Population 2014 | Population 2019 | Population 2024 |
|---|---|---|---|
| Canas | 30,458 | 27,212 | 25,883 |
| Canas Urbano | 17,363 | 15,810 | 15,839 |
| Playa | 13,695 | 11,215 | 11,467 |
| Machuelo Abajo | 10,546 | 10,548 | 10,038 |
| Machuelo Arriba | 11,892 | 11,008 | 10,027 |
| San Antón | 10,882 | 9,888 | 8,314 |
| Coto Laurel | 6,232 | 5,223 | 8,133 |
| Sabanetas | 5,045 | 5,100 | 5,362 |
| Segundo | 7,418 | 4,807 | 4,950 |
| Magueyes | 6,360 | 6,599 | 4,842 |
| Cerrillos | 4,930 | 4,008 | 3,466 |
| Portugués | 4,208 | 3,786 | 3,249 |
| Bucana | 3,618 | 2,625 | 2,755 |
| Primero | 3,071 | 2,654 | 2,512 |
| Portugués Urbano | 3,918 | 3,134 | 2,399 |
| Sexto | 3,670 | 2,676 | 2,297 |
| Real | 3,826 | 2,584 | 2,190 |
| Marueño | 1,337 | 1,611 | 1,223 |
| Cuarto | 1,530 | 1,687 | 1,140 |
| Magueyes Urbano | 1,006 | 1,033 | 1,116 |
| Quebrada Limón | 745 | 435 | 738 |
| Anón | 1,671 | 1,198 | 733 |
| Guaraguao | 1,187 | 488 | 710 |
| Capitanejo | 843 | 936 | 697 |
| Quinto | 451 | 539 | 620 |
| Vayas | 682 | 605 | 599 |
| Tercero | 830 | 539 | 577 |
| Tibes | 1,012 | 523 | 340 |
| Maragüez | 328 | 387 | 190 |

| | | | |
|---|---|---|---|
| Monte Llano | 397 | 391 | 177 |
| San Patricio | 509 | 422 | 177 |

Fuente: US Census ACS

**Diseño de Muestra**

N = 132,760

| Variable | Valor |
|---|---|
| Intervalo de confianza | 90% |
| Desviación estándar | 0.75 |
| Márgen de error | 0.05 |
| Población | 132760 |
| Z-score | 1.645 |
| Muestra infinita | 608.85563 |

| | |
|---|---|
| Muestra | 606.08 |

| Barrio / Sector | Población (2024) | Porciento | Encuestas a realizar |
|---|---|---|---|
| Canas | 25,883 | 19.50% | 118 |
| Canas Urbano | 15,839 | 11.93% | 72 |
| Playa | 11,467 | 8.64% | 52 |
| Machuelo Abajo | 10,038 | 7.56% | 46 |
| Machuelo Arriba | 10,027 | 7.55% | 46 |
| San Antón | 8,314 | 6.26% | 38 |
| Coto Laurel | 8,133 | 6.13% | 37 |
| Sabanetas | 5,362 | 4.04% | 24 |
| Segundo | 4,950 | 3.73% | 23 |
| Magueyes | 4,842 | 3.65% | 22 |
| Cerrillos | 3,466 | 2.61% | 16 |
| Portugués | 3,249 | 2.45% | 15 |
| Bucaná (Buca0) | 2,755 | 2.08% | 13 |
| Primero | 2,512 | 1.89% | 11 |
| Portugués Urbano | 2,399 | 1.81% | 11 |
| Sexto | 2,297 | 1.73% | 10 |
| Real | 2,190 | 1.65% | 10 |
| Marueño | 1,223 | 0.92% | 6 |
| Cuarto | 1,140 | 0.86% | 5 |
| Magueyes Urbano | 1,116 | 0.84% | 5 |
| Quebrada Limón | 738 | 0.56% | 3 |
| Anón | 733 | 0.55% | 3 |
| Guaraguao | 710 | 0.53% | 3 |
| Capitanejo | 697 | 0.52% | 3 |
| Quinto | 620 | 0.47% | 3 |

| | | | |
|---|---|---|---|
| Vayas | 599 | 0.45% | 3 |
| Tercero | 577 | 0.43% | 3 |
| Tibes | 340 | 0.26% | 2 |
| Maragüez | 190 | 0.14% | 1 |
| San Patricio | 177 | 0.13% | 1 |
| Monte Llano | 177 | 0.13% | 1 |
| TOTAL | 132,760 | 100.00% | 606 |

1. **Codigo de Rstudio**

```
#Instalacion de paquetes
install.packages("tidycensus")
install.packages("tidyverse")
install.packages("janitor")
install.packages("purrr")
install.packages("dplyr")
install.packages("tidyr")
install.packages("data.table")

#Activacion de paquetes
```

```
library(tidycensus)
library(tidyverse)
library(janitor)
library(purrr)
library(dplyr)
library(tidyr)
library(data.table)

#Activar API del Censo. Entrar codigo API usando comillas ""
census_api_key("00bd97051ecc23de4c25d76e92c060ab7a358b63", overwrite = TRUE, install
= TRUE)

#Obtención de variables según número de tabla especificado por el Negociado del Censo

vars<- c(
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  Income_15k_24.9k = "B07010_005" ,
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  employment_sex = "B23001_001" )



varstarea2 <- c(
  totpop = "B01001_001",            # Población total
  Median_HH_income = "B19013_001",   # Mediana de ingreso
  Median_age = "B01002_001",         # Mediana de edad
  Pop_poverty_total = "B17001_001",  # Total para calcular % de pobreza
  Pop_under_povln = "B17001_002",    # Población bajo nivel de pobreza
```

```
  Labor_force_total = "B23025_001",  # Población 16+ (Base para calcular empleo)
  Employed = "B23025_004",           # Población empleada
  Unemployed = "B23025_005"          # Población desempleada
)

years<- list(2014,2019,2024)

#Generar tablas

PuertoRicocensus<- map_dfr(
  years,
  ~   get_acs(
    geography = "state",
    state = "PR",
    variables = vars,
    year = .x,
    survey = "acs5",
    output="wide"),
  .id = "year")

PuertoRico_municipio_census<- map_dfr(
  years,
  ~   get_acs(
    geography = "county",
    state = "PR",
    variables = vars,
    year = .x,
    survey = "acs5",
    output="wide"),
  .id = "year")

PuertoRico_tract_census<- map_dfr(
  years,
  ~   get_acs(
    geography = "tract",
    state = "PR",
    variables = vars,
    year = .x,
    survey = "acs5",
    output="wide"),
  .id = "year")

PuertoRico_barrio_census<- map_dfr(
  years,
```

```
  ~   get_acs(
    geography = "county subdivision",
    state = "PR",
    county = "Ponce",
    variables = varstarea2,
    year = .x,
    survey = "acs5",
    output="wide"),
  .id = "year")



PuertoRicocensus$year= as.numeric(PuertoRicocensus$year)+2009
PuertoRico_municipio_census$year= as.numeric(PuertoRico_municipio_census$year)+2009
PuertoRico_tract_census$year= as.numeric(PuertoRico_tract_census$year)+2009
PuertoRico_barrio_census$year= as.numeric(PuertoRico_barrio_census$year)+2009

#Generar archivos en CSV
write.csv(PuertoRicocensus, file = 'PuertoRico_Census.csv')
write.csv(PuertoRico_municipio_census, file = 'PuertoRico_Municipios_Census.csv')
write.csv(PuertoRico_tract_census, file = 'PuertoRico_Tracts_Census.csv')
write.csv(PuertoRico_barrio_census, file = 'PuertoRico_barrio_Census.csv')

variables<-load_variables(year = 2024, "acs5", cache = TRUE)
write.csv(variables, file = "PRCS_2024_Variables.csv")
```